

A Representation-Diagnostic Framework for Arabic Heritage Language Models: A Case Study on Fine-Tuning with the Shamela Corpus

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Abstract

Large language models adapted to specialized historical corpora may develop substantial changes in their internal representations, yet conventional evaluation metrics provide limited information about how such changes evolve across network depth. This paper presents a representation-diagnostic framework for examining layer-wise representation structure in Arabic heritage language models undergoing continued pretraining on the Shamela corpus.

Rather than treating internal representations as direct evidence of causal mechanisms, the proposed framework characterizes them using complementary descriptive diagnostics, including centered kernel alignment (CKA), linear probe performance, macro-F1, silhouette and Davies–Bouldin scores, stable rank, and far-layer representation similarity. We apply the framework to a 4B-parameter Qwen3.5-based model after continued pretraining on six domains of Islamic heritage literature: Aqeedah, Tafsir, Nahw and Sarf, Fiqh, Hadith Mutoon, and Fatawa.

The analysis reveals a pronounced increase in domain discriminability with depth. Linear-probe accuracy rises from 53.80% at layer 0 to 82.31% at the final layer, while macro-F1 increases from 0.537 to 0.824. At the same time, the average cross-domain CKA reaches a maximum of 0.1487 at layer 8 and decreases sharply to 0.0259 at layer 31. Stable rank follows a non-monotonic trajectory, reaching 24.854 at layer 8 before decreasing to 5.762 at the final layer.

These observations indicate that the final representation is substantially more domain-discriminative and considerably lower-rank than intermediate representations. However, these measurements do not by themselves establish knowledge compartmentalization, causal specialization, architectural saturation, or protection against catastrophic forgetting. We therefore interpret the findings as evidence of representational specialization and compression, while identifying several experiments required to determine their functional consequences.

Keywords: Arabic language models; representation analysis; CKA; linear probes; stable rank; Shamela; continued pretraining; heritage Arabic; LLMs; representation diagnostics.

1 Introduction

Large language models have demonstrated increasingly strong capabilities across multilingual and domain-specific applications. However, adapting general-purpose language models to historical and specialized corpora raises an important question: how do internal representations change as the model becomes increasingly exposed to a specialized domain?

This question is particularly relevant for Arabic heritage literature. Classical Islamic texts contain substantial lexical, syntactic, stylistic, and terminological variation. The Shamela collection provides a large body of historical Islamic literature covering multiple disciplines, including jurisprudence, Qur’anic exegesis, Hadith, Arabic grammar, theology, and fatwa literature.

Conventional evaluation based on language-model loss or downstream task accuracy can reveal whether a model improves, but does not directly describe the geometry of its internal

representations. Representation similarity analysis provides one complementary approach. In particular, centered kernel alignment (CKA) has been widely used to compare neural representations across layers and models [Kornblith et al., 2019]. CKA is attractive because it can capture similarity between high-dimensional representations without requiring a direct one-to-one correspondence between individual features.

Another useful perspective is provided by dimensionality measures. Previous work has shown that neural representations can have effective dimensionality far smaller than the nominal number of hidden units, with dimensionality changing substantially across depth [Ansuini et al., 2019]. Such findings motivate examining rank-related statistics in language models rather than interpreting hidden size alone as an indication of effective representational capacity.

The present study therefore focuses on a descriptive question: *How does the representational structure of a Shamela-continued-pretrained Arabic language model change across network depth, and how does this change relate to the discriminability of Islamic textual domains?*

The contribution of this work is a practical diagnostic framework combining several complementary measurements. The framework is intended to help researchers inspect internal representation trajectories before making stronger claims about model scaling, further continued pretraining, or downstream adaptation.

Importantly, this work does not claim to provide mechanistic interpretability in the strict causal sense. The measurements presented here characterize representational geometry and predictive information. They do not identify individual causal circuits, neurons, attention heads, or mechanisms responsible for the observed behavior.

2 Research Questions

The study addresses four research questions.

RQ1: How does domain discriminability change across model depth?

RQ2: How does cross-domain representational similarity change across layers?

RQ3: Does the effective rank of the activation space exhibit a characteristic depth-wise trajectory?

RQ4: Does the final layer exhibit a distinctive combination of domain discriminability, representational similarity, and rank?

These questions are deliberately descriptive. In particular, the study does not attempt to infer from representation statistics alone whether the model has reached an architectural capacity limit.

3 Related Work

3.1 Representation Similarity

Representation similarity analysis has become an important methodology for studying neural networks. CKA was introduced as a robust similarity measure for comparing representations and addressing limitations of conventional CCA-based approaches [Kornblith et al., 2019].

For centered activation matrices X and Y , linear CKA can be expressed as

$$\text{CKA}(X, Y) = \frac{\|X^T Y\|_F^2}{\|X^T X\|_F \|Y^T Y\|_F}. \quad (1)$$

CKA values close to one indicate stronger representational similarity, whereas lower values indicate weaker similarity under the selected representation and sample distribution.

An important methodological point is that low CKA between two domains does not necessarily mean that the corresponding knowledge is independent. Similarity is affected by the activation distribution, preprocessing, sampling strategy, layer location, and the statistical properties of the representation.

3.2 Dimensionality of Neural Representations

Ansuini et al. [Ansuini et al., 2019] showed that the intrinsic dimensionality of neural representations can be substantially smaller than the nominal number of hidden units. Their work also demonstrated non-monotonic dimensionality trajectories across layers.

This motivates the use of rank-based diagnostics in the present study. Stable rank provides a relatively simple measure of the effective spectral spread of an activation matrix.

3.3 Depth Scaling and Continued Pretraining

Depth up-scaling methods such as SOLAR demonstrate that increasing model depth through layer-level transformations followed by continued pretraining can be an effective scaling strategy [Kim et al., 2024]. However, whether such an intervention is appropriate for a particular model cannot be inferred from a single representation statistic.

The present work therefore treats architectural expansion as a possible future intervention rather than as a conclusion derived directly from CKA or stable rank.

4 Methodology

4.1 Framework Overview

The diagnostic framework contains four principal analytical components:

1. domain-level CKA;
2. linear-probe domain classification;
3. layer-wise stable-rank analysis;
4. far-layer CKA analysis.

Together, these measurements provide complementary information about representational similarity, domain discriminability, and spectral structure.

4.2 Centered Kernel Alignment

For two centered activation matrices $X, Y \in \mathbb{R}^{n \times d}$, linear CKA is calculated as

$$\text{CKA}(X, Y) = \frac{\|X^T Y\|_F^2}{\|X^T X\|_F \|Y^T Y\|_F}. \quad (2)$$

For cross-domain analysis, the activations of two different textual domains are compared at the same model layer.

The resulting quantity should be interpreted as a measure of representational similarity rather than as a direct measure of semantic independence or knowledge separation.

4.3 Linear Probe

A linear classifier is trained on the activation vectors to predict the source domain of each text.

Six classes are used: Aqeedah; Tafsir; Nahw and Sarf; General Fiqh; Hadith Mutoon; Fatawa.

Performance is reported using accuracy and macro-F1.

A high linear-probe score indicates that domain-discriminative information is accessible through a linear transformation of the representation. It does not imply that the model explicitly contains a symbolic domain classifier.

4.4 Stable Rank

For an activation matrix A with singular values $\sigma_1, \dots, \sigma_r$, stable rank is defined as

$$\text{StableRank}(A) = \frac{\|A\|_F^2}{\|A\|_2^2} = \frac{\sum_i \sigma_i^2}{\sigma_{\max}^2}. \quad (3)$$

Stable rank is bounded above by the ordinary matrix rank and provides a spectral measure of effective dimensionality.

In this study, stable rank is reported both in absolute terms and relative to the hidden dimension of 2560.

A low stable rank should not automatically be interpreted as either efficient representation or insufficient model capacity. Both interpretations require additional evidence.

4.5 Far-Layer CKA

Far-layer CKA measures similarity between representations from layers that are separated by substantial network depth.

The analysis includes:

- L00 $\leftrightarrow$ L16;
- L00 $\leftrightarrow$ L31;
- L04 $\leftrightarrow$ L28;
- L08 $\leftrightarrow$ L31.

The L05 $\leftrightarrow$ L25 entries in the raw diagnostic output are zero because those layer activations were not collected in the V10 layer set. They are therefore treated as unavailable rather than as genuine evidence of zero similarity.

5 Experimental Setup

5.1 Model

The evaluated model is a Qwen3.5-based 4B-parameter language model continued-pretrained on the Shamela corpus.

The architecture used in the diagnostic run has:

- hidden dimension: 2560;
- number of layers: 32;
- diagnostic layers: $\{0, 4, 8, 12, 16, 20, 24, 28, 31\}$;
- final diagnostic layer: L31.

The publicly documented Qwen3.5-4B architecture specifies a 2560-dimensional hidden representation and 32 layers, with its attention configuration containing 16 query attention heads in the gated-attention component.

5.2 Continued Pretraining Data

Six domains from the Shamela collection were used:

Table 1: Domains used in the representation analysis.

Domain	Texts available
Aqeedah	100,000
Tafsir	100,000
Nahw and Sarf	100,000
General Fiqh	100,000
Hadith Mutoon	100,000
Fatawa	100,000
Total	600,000

The diagnostic collection itself used 800 activation vectors per domain and layer.

Thus, for every domain-layer pair, the activation matrix had dimensions 800×2560 .

The activation vector corresponds to the selected final real token after attention processing, according to the V10 collection procedure.

5.3 Activation Collection

Activations were collected from nine layers: $L = \{0, 4, 8, 12, 16, 20, 24, 28, 31\}$.

For each of the six domains, 800 vectors were collected at each diagnostic layer.

Consequently, the complete collection contained $6 \times 9 \times 800 = 43,200$ activation vectors.

The resulting matrices were retained separately by domain and layer to permit both within-layer cross-domain analysis and within-domain cross-layer analysis.

5.4 Hardware and Inference

The V10 analysis was executed on a single NVIDIA RTX 3090 GPU with 24 GB of VRAM.

During the run, approximately 7.8 GB of GPU memory was allocated by the diagnostic process.

Activation collection completed successfully over 1200 collection batches.

6 Results

6.1 Layer-wise Domain Discriminability

The strongest result is the progressive increase in linear-probe domain-discriminability across depth.

Table 2: Linear-probe performance across diagnostic layers.

Layer	Accuracy	Macro-F1	Silhouette	Davies-Bouldin
0	53.80%	0.537	-0.0075	13.5336
4	70.28%	0.703	-0.0060	11.1941
8	72.87%	0.728	-0.0112	11.2660
12	76.76%	0.766	-0.0089	11.0906
16	78.06%	0.782	-0.0068	10.1647
20	79.54%	0.797	-0.0029	8.1488
24	79.72%	0.799	-0.0110	8.3116
28	77.69%	0.777	-0.0124	8.4527
31	82.31%	0.824	0.0031	6.7378

Accuracy increases from 53.80% at layer 0 to 82.31% at layer 31. Macro-F1 follows a similar trajectory, increasing from 0.537 to 0.824.

Interestingly, the progression is not strictly monotonic. Performance reaches 79.72% at L24, decreases to 77.69% at L28, and then increases to 82.31% at L31.

This suggests that the final layer contains particularly accessible domain-discriminative information, but the non-monotonic behavior indicates that representation development is more complex than a simple linear increase in specialization.

6.2 Cross-Domain CKA at the Final Layer

Table 3 reports the cross-domain CKA matrix at L31.

Table 3: Cross-domain CKA matrix at layer 31.

	Aqeedah	Tafsir	Nahw	Fiqh	Mutoon	Fatawa
Aqeedah	1.0000	0.0276	0.0258	0.0262	0.0312	0.0375
Tafsir	0.0276	1.0000	0.0228	0.0172	0.0293	0.0217
Nahw	0.0258	0.0228	1.0000	0.0173	0.0254	0.0258
Fiqh	0.0262	0.0172	0.0173	1.0000	0.0213	0.0222
Mutoon	0.0312	0.0293	0.0254	0.0213	1.0000	0.0371
Fatawa	0.0375	0.0217	0.0258	0.0222	0.0371	1.0000

All off-diagonal CKA values are below 0.04.

These low values indicate that the activation distributions associated with the six domains are substantially dissimilar at L31 under the present CKA measurement.

However, low CKA should not be interpreted as proof that the model stores knowledge in independent compartments. CKA measures representational similarity, not causal independence or resistance to interference.

6.3 Average Domain CKA Across Depth

A particularly notable result is the non-monotonic evolution of average cross-domain CKA.

Table 4: Average cross-domain CKA by layer.

Layer	Average Domain CKA
0	0.054200
4	0.127515
8	0.148687
12	0.146753
16	0.130844
20	0.112298
24	0.116429
28	0.117917
31	0.025891

Average domain CKA rises from 0.0542 at L0 to a maximum of 0.1487 at L8. It then gradually decreases and reaches 0.025891 at L31.

The sharp reduction between L28 and L31 is particularly notable because it occurs simultaneously with the highest linear-probe accuracy.

One possible descriptive interpretation is that the final representation makes domain-discriminative information more accessible while becoming less similar across domains. However, establishing whether this transition is functionally necessary would require intervention experiments.

6.4 Stable Rank

The stable-rank trajectory is shown in Table 5.

Table 5: Layer-wise stable rank of activation representations.

Layer	Stable Rank	Stable Rank / 2560
0	10.593	0.004138
4	20.324	0.007939
8	24.854	0.009709
12	23.687	0.009253
16	18.208	0.007113
20	16.453	0.006427
24	20.457	0.007991
28	18.448	0.007206
31	5.762	0.002251

Stable rank increases substantially during the early layers, reaching 24.854 at L8. It subsequently decreases, with a pronounced reduction at the final layer.

The L31 value of 5.762 is substantially lower than the nominal hidden dimension of 2560.

This result should be interpreted cautiously. Stable rank captures spectral structure, but it cannot by itself distinguish between efficient compression, task-specific specialization, anisotropy, or under-utilization of representational dimensions.

6.5 Far-Layer CKA

Far-layer CKA provides additional information about the evolution of representations within each domain.

Table 6: Far-layer CKA for the six domains.

Domain	L00–L16	L00–L31	L04–L28	L08–L31
Aqeedah	0.58293	0.34888	0.59194	0.43168
Tafsir	0.52570	0.39176	0.70091	0.45480
Nahw/Sarf	0.53741	0.37435	0.65235	0.50708
Fiqh	0.86112	0.49430	0.90041	0.57715
Mutoon	0.55544	0.34209	0.65623	0.41128
Fatawa	0.75068	0.58618	0.70106	0.57495

The results show substantial within-domain similarity across distant layers. For example, Fiqh reaches 0.90041 for L04–L28, while Fatawa reaches 0.75068 for L00–L16.

Importantly, these results also demonstrate why a single far-layer CKA value should not be used to characterize the entire network. Similarity depends on the selected layers and domain.

The raw V10 output contains an L05–L25 value of 0.00000 for every domain. Because L5 and L25 were not included in the activation collection set, these values represent unavailable measurements rather than observed zero similarity. They are therefore excluded from substantive interpretation.

6.6 Integrated Layer Analysis

Figure 1 summarizes the principal V10 measurements.

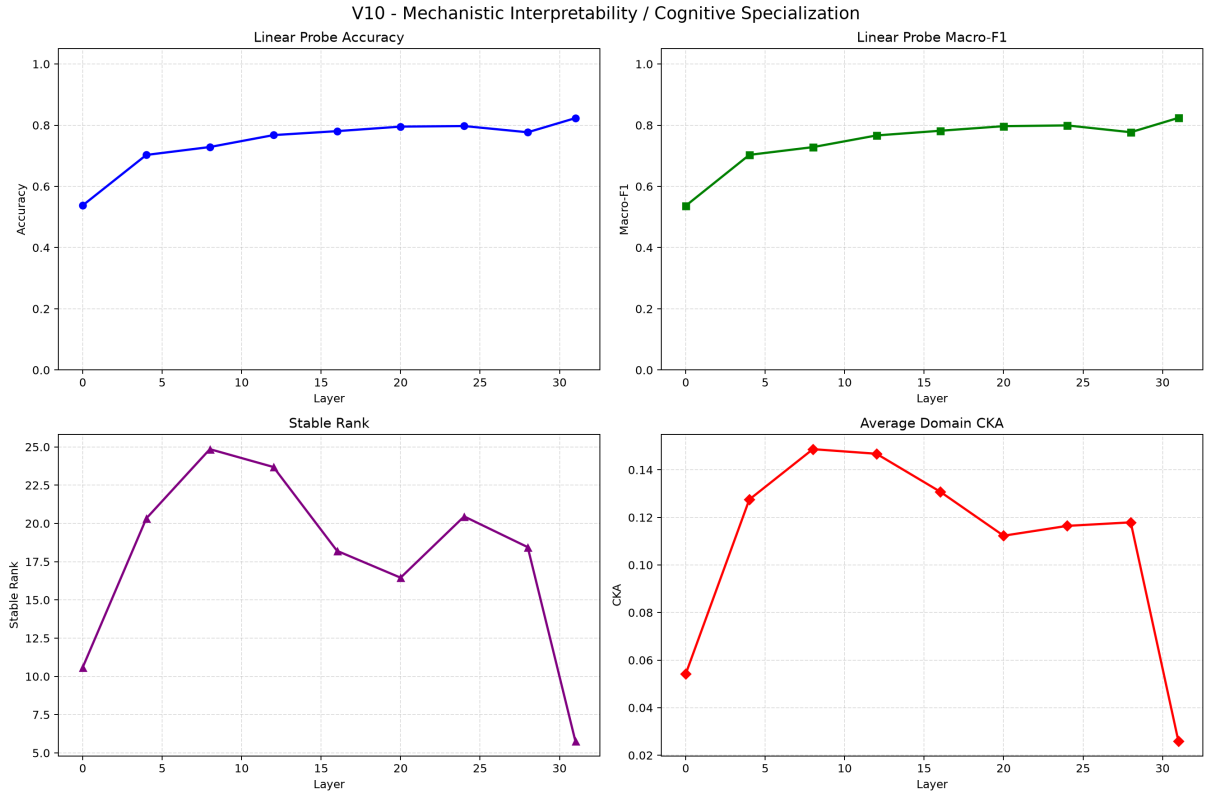


Figure 1: V10 representation diagnostics across model depth. The panels show linear-probe accuracy, macro-F1, stable rank, and average cross-domain CKA. The final layer exhibits the highest domain-probe performance, the lowest stable rank, and the lowest average cross-domain CKA.

The joint pattern is more informative than any single metric.

At intermediate layers, stable rank is comparatively higher and cross-domain CKA is also higher. At L31, domain discriminability reaches its maximum while both stable rank and cross-domain CKA reach pronounced minima.

This convergence of three measurements makes the final layer particularly interesting for further study.

7 Discussion

7.1 Increasing Domain Discriminability

The clearest empirical observation is that domain identity becomes increasingly linearly decodable with depth.

The increase from 53.80% at L0 to 82.31% at L31 indicates that the final representations contain substantially more linearly accessible information about the source discipline.

This is consistent with the general observation that deeper neural representations can become increasingly specialized for task-relevant information.

Nevertheless, domain classification accuracy should not be equated with “knowledge separation.” A representation may encode domain identity for many reasons, including lexical distributions, stylistic patterns, document structure, and semantic content.

Therefore, the present result is best described as: *The model’s final-layer representations contain strongly domain-discriminative information.*

This statement is directly supported by the linear-probe results without requiring assumptions about the internal organization of knowledge.

7.2 The Final-Layer Compression Pattern

The most interesting structural observation is the simultaneous occurrence of:

1. maximum linear-probe accuracy;
2. minimum average cross-domain CKA;
3. very low stable rank.

At L31, the model reaches 82.31% linear-probe accuracy while stable rank falls to 5.762 and average cross-domain CKA falls to 0.025891.

One hypothesis is that the final representation compresses information into a smaller number of directions that remain highly discriminative for the domains represented in the training data.

However, this remains a hypothesis.

An equally plausible alternative is that the final layer produces an anisotropic representation in which only a small number of directions dominate the variance.

Distinguishing these interpretations requires additional measurements, including covariance spectra, participation ratio, effective rank, intrinsic dimension, and comparisons against the original base model.

7.3 Why Low CKA Does Not Establish Knowledge Compartmentalization

The low cross-domain CKA values are striking, but they should not be interpreted as evidence that knowledge is stored in independent modules.

If two domains have different lexical, stylistic, and semantic distributions, their activations can naturally become dissimilar even when they share large amounts of underlying knowledge.

Furthermore, CKA is a statistical similarity measure. It does not provide a causal intervention.

Consequently, the present results support the weaker and more defensible claim that: *The model produces substantially different activation patterns for the six textual domains at its final layer.*

Whether those differences reduce interference during subsequent training is an open empirical question.

7.4 Implications for Continued Pretraining

The results do not provide sufficient evidence to claim that the model has reached an architectural saturation point.

In particular, stable rank of 5.762 does not imply that the model has “used only 5.762 dimensions” of its 2560-dimensional hidden state in a literal sense.

Likewise, low CKA does not establish that adding layers would be unnecessary.

Architectural decisions should instead be informed by a combination of:

- validation loss;
- per-domain validation loss;
- downstream task performance;
- representation trajectories across checkpoints;
- probe performance;
- forgetting measurements;
- comparisons with the original base model.

Depth up-scaling is therefore best considered a future experimental condition rather than a recommendation derived from the current diagnostics.

8 Limitations

This study has several important limitations.

8.1 No Base-Model Comparison

The current analysis focuses on the continued-pretrained checkpoint and does not include the original Qwen3.5-4B base model under identical diagnostic conditions.

Therefore, we cannot determine which representational changes are caused specifically by continued pretraining on Shamela.

A direct base-versus-continued-pretraining comparison is necessary.

8.2 Single Main Checkpoint

The analysis focuses on checkpoint 97,500.

Although intermediate layers within this checkpoint were analyzed, the study does not yet provide a trajectory across training checkpoints.

Consequently, the designation of a “saturation point” cannot be established from these data.

Future experiments should analyze checkpoints such as 10k, 30k, 50k, 75k, and 97.5k steps.

8.3 No Statistical Confidence Intervals

The reported linear-probe accuracy of 82.31% is based on a single diagnostic run.

Confidence intervals, bootstrap estimates, repeated splits, and multiple random seeds should be reported in future experiments.

8.4 Representation Analysis Is Not Mechanistic Interpretability

The current methodology analyzes representation geometry and predictive information.

It does not identify causal mechanisms, circuits, neurons, attention heads, or interventions that change model behavior.

For this reason, the paper deliberately uses the term *representation-diagnostic* rather than claiming mechanistic interpretability in the strict sense.

8.5 No Direct Catastrophic-Forgetting Experiment

The current experiments do not test whether subsequent supervised fine-tuning would cause catastrophic forgetting.

Low cross-domain CKA and high domain-probe accuracy cannot establish this.

A proper forgetting experiment would require domain-specific evaluation before and after SFT or additional continued pretraining.

9 Future Work

Several experiments follow naturally from the present findings.

9.1 Checkpoint Trajectory

Future work will measure the same metrics across multiple checkpoints: 10k, 30k, 50k, 75k, 97.5k.

This would determine whether the final-layer rank collapse and CKA reduction emerge progressively or occur only at a late stage.

9.2 Base-Model Comparison

The same diagnostic pipeline should be applied to the original Qwen3.5-4B model using matched input samples.

The difference between base and continued-pretrained representations would provide a more direct estimate of the effect of Shamela training.

9.3 Intrinsic Dimension

The current V10 analysis emphasizes stable rank. Future work should add Levina–Bickel maximum-likelihood intrinsic-dimension estimation.

For k nearest neighbors, the estimator can be expressed as $\hat{d} = \frac{k-1}{\sum_{i=1}^{k-1} \log(r_k/r_i)}$.

This would provide a complementary nonlinear estimate of representation dimensionality.

9.4 Effective Rank and Participation Ratio

Additional spectral measures should be computed to determine whether the stable-rank collapse corresponds to broader spectral concentration.

In particular:

$$\text{EffectiveRank}(A) = \exp \left(- \sum_i p_i \log p_i \right), \quad (4)$$

where $p_i = \frac{\sigma_i}{\sum_j \sigma_j}$.

Participation ratio can also be evaluated as

$$PR(A) = \frac{(\sum_i \sigma_i)^2}{\sum_i \sigma_i^2}. \quad (5)$$

9.5 SFT and Forgetting Experiments

The current representation results motivate, but do not establish, an interesting downstream hypothesis: Domain-discriminative representations may influence the extent to which domain-specific SFT causes interference with other domains.

This hypothesis should be tested experimentally.

Performance should be measured on every domain before and after SFT, allowing a direct forgetting matrix to be constructed.

9.6 Causal Representation Interventions

A stronger mechanistic interpretation would require interventions such as activation patching, representation ablation, feature steering, or layer-specific causal mediation experiments.

Such experiments are outside the scope of the present study.

10 Conclusion

We presented a representation-diagnostic framework for studying how internal representations evolve in an Arabic heritage language model continued-pretrained on the Shamela corpus.

The V10 analysis reveals three principal observations.

First, domain discriminability increases substantially with depth. Linear probe accuracy rises from 53.80% at L0 to 82.31% at L31, while macro-F1 increases from 0.537 to 0.824.

Second, cross-domain representational similarity decreases strongly in the final layer. Average domain CKA falls from values around 0.11–0.15 in the middle layers to 0.025891 at L31.

Third, stable rank exhibits a non-monotonic trajectory and reaches 5.762 at the final layer, substantially below the nominal hidden dimension of 2560.

The combination of these results suggests that the final layer has a distinctive representational regime characterized by high domain discriminability and strong spectral compression.

However, these observations should not be interpreted as proof of causal knowledge compartmentalization, architectural saturation, or protection against catastrophic forgetting.

The principal contribution of this work is therefore methodological: representation diagnostics can reveal meaningful changes in the geometry and domain accessibility of Arabic heritage language model representations, while also identifying questions that cannot be answered without additional experiments.

Future work should focus on checkpoint trajectories, base-model comparisons, statistical uncertainty, intrinsic-dimensionality analysis, and direct forgetting experiments. These extensions will determine whether the representation patterns observed here correspond to functional advantages during subsequent training and downstream adaptation.

Data and Code Availability

The diagnostic implementation and model artifacts are associated with the Shamela Builder Project.

The continued-pretrained model checkpoint analyzed in this study is: <https://huggingface.co/sherif1313/3arabLM-4B-Fiqh-v1>

The diagnostic outputs include:

- `linear_probe_results.csv`;
- `stable_rank_results.csv`;
- `domain_cka_results.csv`;

- `far_cka.csv`;
- `combined_layer_analysis.csv`;
- `layer_analysis_v10.png`.

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